```
!pip install -q sentence-transformers scikit-learn
```

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```

```
import pandas as pd
import numpy as np
from sentence_transformers import SentenceTransformer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("/content/cleaned_customer_support_english.csv")
df.head(5)
```

	cleaned_text	type	priority	queue
0	account disruption dear customer support teamn...	Incident	high	Technical Support
1	query smart home system integration features d...	Request	medium	Returns and Exchanges
2	inquiry regarding invoice details dear custome...	Request	low	Billing and Payments
3	question marketing agency software compatibili...	Problem	medium	Sales and Pre-Sales
4	feature query dear customer supportnni hope me...	Request	high	Technical Support

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['type'].value_counts()
```

type	count
Incident	9519
Request	6902
Problem	4984
Change	2535

dtype: int64

```
texts=df['cleaned_text'].tolist()
labels=df['type'].tolist()
```

```
print("🔄 Generating embeddings...")
model = SentenceTransformer('all-mpnet-base-v2')
embeddings = model.encode(texts, show_progress_bar=True)
```

```
🔄 Generating embeddings...
Batches: 100%  749/749 [01:16<00:00, 33.64it/s]
```

```
X = np.array(embeddings)
y = labels

X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42)
```

```
print(" Training base model...")
clf = LogisticRegression(max_iter=5000, class_weight='balanced')
clf.fit(X_train, y_train)
```

```
Training base model...
▾ LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=5000)
```

```
y_pred = clf.predict(X_test)
```

```
print("\n Accuracy:", accuracy_score(y_test, y_pred))
print("\n Classification Report:\n", classification_report(y_test, y_pred))
```

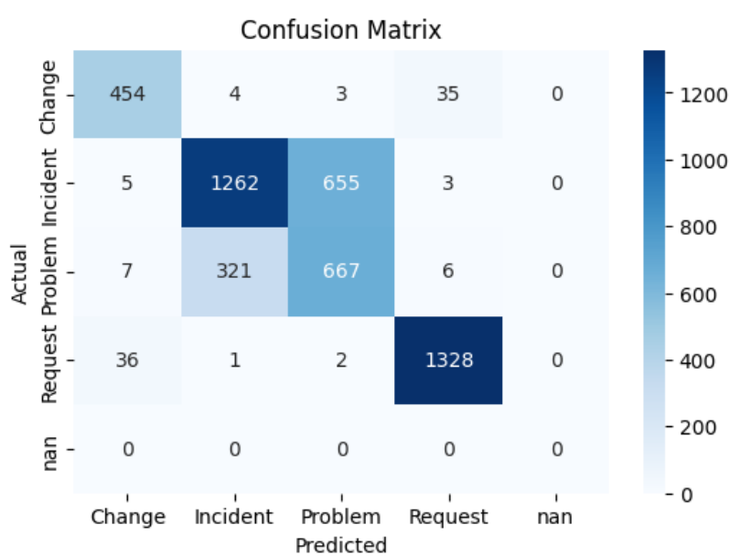
```
 Accuracy: 0.774900814366256

 Classification Report:
               precision    recall  f1-score   support

       Change       0.90      0.92      0.91       496
     Incident       0.79      0.66      0.72      1925
      Problem       0.50      0.67      0.57      1001
      Request       0.97      0.97      0.97      1367

     accuracy                           0.77      4789
    macro avg       0.79      0.80      0.79      4789
weighted avg       0.79      0.77      0.78      4789
```

```
cm = confusion_matrix(y_test, y_pred, labels=clf.classes_)
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=clf.classes_, yticklabels=clf.classes_)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```




```
from sklearn.svm import SVC
print("Training SVM...")
svm_clf = SVC(kernel='linear', class_weight='balanced')
svm_clf.fit(X_train, y_train)
```

```
Training SVM...
SVC
SVC(class_weight='balanced', kernel='linear')
```

```
y_pred_svm = svm_clf.predict(X_test)
```

```
print("\n Accuracy:", accuracy_score(y_test, y_pred_svm))
print("\n Classification Report:\n", classification_report(y_test, y_pred_svm))
```

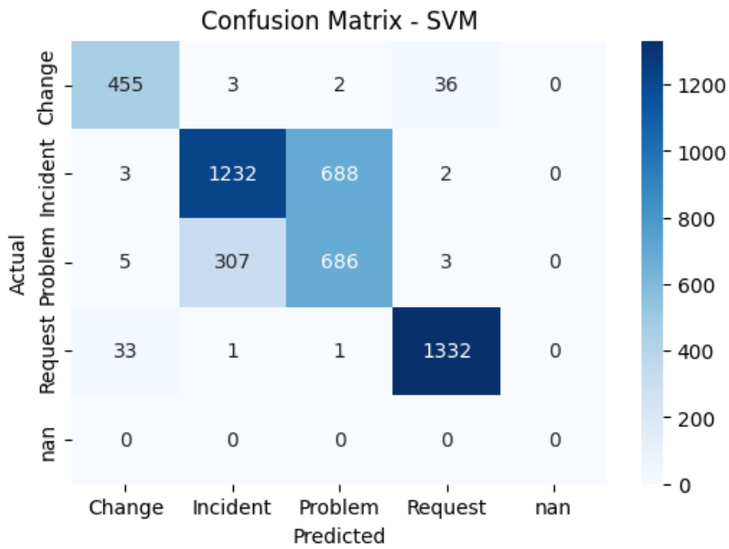
```
 Accuracy: 0.773647943203174

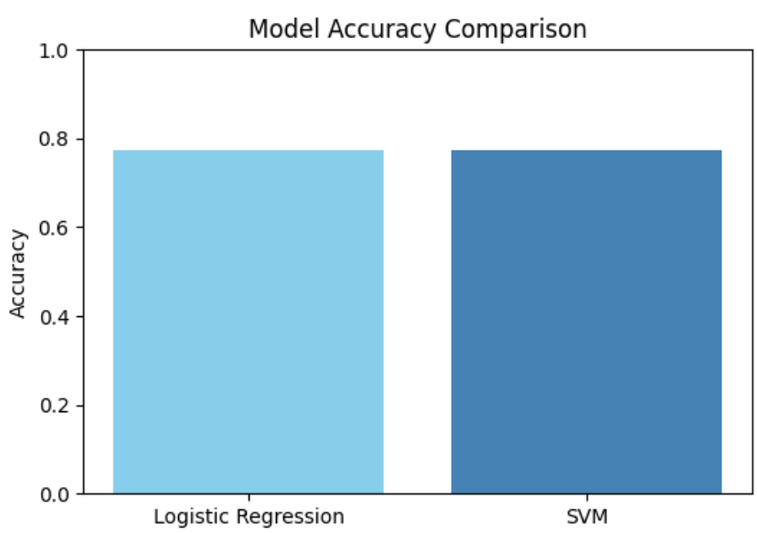
 Classification Report:
               precision    recall  f1-score   support

       Change       0.92      0.92      0.92       496
     Incident       0.80      0.64      0.71      1925
      Problem       0.50      0.69      0.58      1001
      Request       0.97      0.97      0.97      1367

     accuracy                           0.77      4789
    macro avg       0.80      0.80      0.79      4789
weighted avg       0.80      0.77      0.78      4789
```

```
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred_svm, labels=svm_clf.classes_)
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=svm_clf.classes_, yticklabels=svm_clf.classes_)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix - SVM")
plt.show()
```




```
models = ['Logistic Regression', 'SVM']
accuracies = [0.7749, 0.7736]

plt.figure(figsize=(6,4))
plt.bar(models, accuracies, color=['skyblue','steelblue'])
plt.ylabel("Accuracy")
plt.ylim(0,1)
plt.title("Model Accuracy Comparison")
plt.show()
```